

Resolved Neutral Hydrogen in an Infalling Galaxy Population: a 100-hour SKA-Mid Pathfinder Program

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Requested telescope: SKA-Mid • Requested time: 100 hours • Array: AA*, Band 2

Abstract

We request 100 hours with SKA-Mid to obtain resolved ($\approx 15''$, ≈ 2.8 km/s) 21-cm HI imaging of twelve galaxies spanning the outskirts and infall region of a nearby ($D \approx 20$ Mpc), low-mass galaxy cluster. Neutral hydrogen is the most extended and most fragile baryonic component of a galaxy, and is therefore the first casualty of environmental processing as galaxies fall into groups and clusters. Deep, spatially resolved HI mapping is required to trace the tidal tails, warps, truncated discs and one-sided ‘jellyfish’ morphologies produced by tidal interactions and ram-pressure stripping, and to measure how these mechanisms shape the atomic gas reservoir before it is transformed into stars or removed altogether. Existing precursor and pathfinder surveys (MeerKAT, ASKAP, WSRT) have identified a first generation of such features around low-mass clusters, but are sensitivity- and resolution-limited outside the very nearest systems. SKA-Mid AA* — 144 dishes, comprising 80 SKA-Mid 15-m dishes and the 64-dish MeerKAT array, together with the SKA1 core baselines — will reach a 3σ column density of $\approx 2.1 \times 10^{19} \text{ cm}^{-2}$ at $15'' \times 13''$ resolution and 8.3 km/s velocity resolution for our sample in a modest 6 hours per target: several times sharper, and comparably deep, to what precursor instruments deliver in similar time, and achievable with the array configuration expected to be available ahead of the completed AA4. This program is deliberately scoped as a small, single-semester pathfinder: it targets a well-defined, previously identified sample rather than a blind survey, and is designed to validate the SKA-Mid HI environmental-science pipeline ahead of larger Key Science Programmes.

1. Scientific Justification

1.1 HI as a tracer of the baryon cycle and environmental processing

Neutral atomic hydrogen (HI) is the raw material from which molecular gas and stars form, and its abundance, distribution and kinematics record the full baryon cycle of accretion, star formation and feedback that drives galaxy evolution [1,2]. Because HI discs extend well beyond a galaxy's stellar body and are only weakly self-gravitating in their outskirts, HI is the most sensitive tracer of external perturbations: gravitational tides, harassment, and the ram pressure exerted by a hot intracluster or intragroup medium as a galaxy falls inward [3,4]. Deep HI imaging of galaxy groups and cluster outskirts over the last decade — with the WSRT, JVLA, ASKAP and MeerKAT — has revealed an increasingly rich population of asymmetric HI discs, extraplanar gas, tidal tails and bridges, polar and warped structures, and one-sided ‘jellyfish’ tails, and has shown that these features occur not only in massive clusters but also in lower-mass groups and cosmic-web filaments [3].

A particularly instructive laboratory is the outskirts and infall region of nearby, low-mass clusters, where deep MeerKAT and other precursor HI imaging has already uncovered long HI tails with mixed tidal-and-ram-pressure origins, and tidal bridges connecting infalling galaxies to their host system [3,5]. These discoveries demonstrate that gas removal begins well before galaxies reach the cluster core, but the existing observations are sensitivity-limited: many of the faintest, most diffuse HI features — which trace the earliest and most informative stages of stripping — remain below current detection thresholds, and the angular resolution achieved to date is often too coarse to disentangle tidal from hydrodynamical signatures or to connect the disturbed gas to the underlying stellar disc.

1.2 Why SKA-Mid, and why now

SKA-Mid AA* (144 dishes: 80 newly-built SKA-Mid 15-m dishes plus the 64-dish MeerKAT array) already delivers both the surface-brightness sensitivity and the angular resolution that this science requires simultaneously — a combination existing telescopes cannot match, even before the array reaches its full AA4 design baseline. Restricting the array to its dense core (baselines ≤ 2 km) recovers most of the AA* collecting area while producing a synthesized beam of order $15''$, well matched to resolving individual tidal features and gas tails in a $D \approx 20$ Mpc system, while the 144-dish collecting area keeps the surface-brightness sensitivity high, at only a modest cost relative to the eventual full AA4 array [6,7]. This chapter of SKA HI science — resolved imaging of environmental gas dynamics — is explicitly identified as one of the flagship applications of SKA-Mid Band 2 HI observations [3,4], complementing the parallel, much larger statistical and cosmological HI surveys (HI mass function evolution, intensity mapping, HI absorption) that require far greater investments of time [2,8].

Because this proposal targets a modest, well-defined sample rather than a blind area survey, it is deliberately scoped to be achievable as a single-cycle, 100-hour pathfinder program: a scale at which SKA-Mid can already produce a scientific advance over precursor data, while also serving as a technical proving ground (RFI excision, joint deconvolution of multiple sessions, and HI source-finding pipelines) ahead of any future, much larger Key Science Programme allocation for HI environmental studies [2].

1.3 Key science questions

- What fraction of the atomic gas in cluster-infall galaxies has already been perturbed or removed before first pericentric passage, and what does the morphology of the disturbed gas (tails, warps, truncation, asymmetry) reveal about the relative roles of tidal interaction versus ram-pressure stripping [3,4]?
- How does the HI morphology and content depend on a galaxy's position within the group/filament/cluster structure, and on its stellar and HI mass [3,9]?
- For the subset of targets with regular, well-resolved discs, how do the outer, dark-matter-dominated regions of their rotation curves compare with isolated field galaxies of similar mass, and are there systematic differences attributable to environmental effects on the baryonic distribution [10]?

2. Immediate Objective and Target Sample

The immediate objective is to obtain resolved HI imaging, at column density and velocity sensitivity well beyond current precursor data, for a sample of 12 galaxies distributed across the outskirts, filament and infall region of a nearby, low-mass, well-studied cluster (illustrative case: the Fornax cluster and its surrounding large-scale structure, $D \approx 20$ Mpc). The sample is drawn directly from existing HI precursor catalogues (e.g. MeerKAT/MHONGOOSE- and ASKAP/WALLABY-type surveys) and targeted follow-up studies that have already flagged candidate tidal and ram-pressure-stripped systems in this environment [3,5], ensuring that every pointing has a well-motivated, pre-identified target rather than relying on a blind search. The sample spans a range of projected clustercentric distance (from the virial radius out into the connecting filament), stellar mass, and HI morphology (regular discs, mildly disturbed discs, and confirmed or candidate jellyfish/tailed systems), so that the trends of gas disturbance with environment and mass can be assessed differentially within a single, homogeneously observed dataset.

Each of the 12 targets is allocated one pointing, with the Band 2 primary beam (≈ 1 degree FWHM for a 15-m SKA1 dish) comfortably encompassing both the stellar disc and its surrounding circumgalactic HI out to several tens of kpc — the scale over which tidal tails and stripped tails are typically observed to extend [3].

3. Description of the Observations

3.1 Array configuration and spectral setup

We request SKA-Mid AA*, Band 2 (0.95–1.76 GHz), tuned to the local recession-velocity range of the target structure (illustrative centre 1413.3 MHz, $cz \approx 1500$ km/s). A single spectral zoom window of 4 MHz (≈ 850 km/s) is centred on each target's systemic velocity, comfortably covering its full HI line width plus surrounding baseline for continuum subtraction; the native 13.4 kHz channels (≈ 2.8 km/s) resolve the line profile finely enough to study non-circular motions in the disturbed gas, while allowing post-processing smoothing to ≈ 8 km/s to recover extended, low column density emission. Robust weighting (robust = 0) applied to the SKA1 core sub-array (baselines ≤ 2 km) delivers a synthesized beam of $\approx 15.2'' \times 12.6''$, matched to the physical scale (a few kpc) of the tidal and stripping features this program is designed to resolve, while retaining most of AA*'s collecting area for sensitivity. Table 1 summarises the observational setup and time request; Table 2 summarises the resulting sensitivity, both derived using the SKAO Sensitivity Calculator [11].

Parameter	Value
Telescope / array	SKA-Mid, Array Assembly * (AA*); 144 dishes (80 SKA-Mid 15-m + 64 MeerKAT 13.5-m), core sub-array restricted to baselines ≤ 2 km
Receiver band	Band 2 (0.95–1.76 GHz)
Tuning	Centred at 1413.3 MHz ($cz \approx 1500$ km/s, matched to the Fornax infall region)
Spectral setup	Zoom window, 4 MHz bandwidth (≈ 850 km/s), 13.4 kHz channels (≈ 2.8 km/s)
Weighting / resolution	Robust (robust = 0), core-restricted array $\rightarrow$ synthesized beam $\approx 15.2'' \times 12.6''$
Targets	12 group/cluster-infall galaxies in the Fornax outskirts and associated filament
On-source time per target	6.0 h
Total on-source time	72.0 h
Calibration & overheads	≈ 22.0 h (bandpass/flux/phase calibration, pointing, slew, RFI margin)
Total time requested	94.0 h (rounded to 100 h)

3.2 Sensitivity

With 6 hours on-source per target, the SKAO Sensitivity Calculator predicts a native-channel (2.8 km/s) rms of $254 \mu\text{Jy/beam}$ at the adopted $\approx 15.2'' \times 12.6''$ resolution; smoothing to 8.3 km/s (a factor-of-3 channel average, well matched to the intrinsic linewidths of the disturbed, extraplanar HI this program targets) yields a 3σ column density sensitivity of $\approx 2.1 \times 10^{19} \text{ cm}^{-2}$ — comparable to the deepest existing precursor HI imaging of cluster environments, and achieved here at higher angular resolution, despite AA* fielding 53 fewer dishes than the eventual AA4 array. For a compact, marginally resolved feature with a 20 km/s line width, the corresponding 5σ integrated HI mass limit at $D = 20$ Mpc is $\approx 9 \times 10^5 M_\odot$ — sufficient to detect not only the disturbed discs of the main targets but also faint, isolated HI clumps stripped from them. These figures are representative order-of-magnitude estimates from the calculator for natural sky and system conditions; final depths will depend on the true weather, elevation and RFI conditions at the time of observation, and would improve further should the program be scheduled after the array grows towards its full AA4 design baseline.

Quantity	Value
Native channel rms (2.8 km/s, 6 h)	254 μJy/beam
Smoothed rms (8.3 km/s, 3-channel average)	≈ 147 μJy/beam
3 σ HI column density (15.2'' $\times$ 12.6'' beam, 8.3 km/s)	$\approx 2.1 \times 10^{19}$ cm^{-2}
5 σ integrated HI mass limit (unresolved, 20 km/s line, D = 20 Mpc)	$\approx 9 \times 10^5$ M$\odot$
Surface-brightness sensitivity (natural units)	≈ 0.81 K (per native channel)

3.3 Time justification

Twelve targets at 6 hours on-source each total 72 hours. We add approximately 22 hours ($\approx 30\%$) for standard calibration overheads — bandpass and flux calibrator scans, phase referencing, pointing checks, slewing between targets, and a margin for loss of frequency channels to radio-frequency interference (RFI), which is a well-documented and unavoidable cost of Band 2 spectral-line observing even at SKA's radio-quiet sites [2]. This gives a total request of 94 hours, which we round to 100 hours to provide a modest contingency for repeat scans on any target affected by unexpectedly poor conditions or additional RFI flagging, consistent with the tiered survey strategies described for SKA-Mid HI programs [2]. We consider six hours per target to be close to the minimum integration that resolves the key diagnostic scale (a few kpc, tens of arcsec) at the column density needed to trace the faint, extended tidal and stripped features that motivate this program; shorter integrations would recover only the brighter, already-known disc emission and would not add resolution or depth beyond existing precursor data.

4. Data Reduction and Analysis Plan

- Standard SKA Science Data Processor calibration and imaging pipelines will be used for bandpass, flux and phase calibration, RFI flagging, and joint deconvolution of the visibilities for each target [2,6].
- HI moment maps (integrated intensity, velocity field, velocity dispersion) will be produced at both native and smoothed spectral resolution, following standard practice for resolved HI kinematic studies [10].
- Morphological and kinematic classification (regular disc, warp, truncation, tail, asymmetry index) will be carried out for each target and compared against clustercentric distance and stellar/HI mass, following the environmental diagnostics developed for precursor HI surveys of clusters and groups [3,4].
- For targets with sufficiently regular kinematics, rotation curves will be extracted and compared with the mass-matched field-galaxy population to search for environmentally-driven differences in the outer, dark-matter-dominated regions [10].
- All calibrated visibilities, images and derived data products will be archived and made available through the SKA Regional Centre Network, consistent with SKAO data policy [12].

5. Feasibility and Risk

- RFI: Band 2 is affected by satellite and ground-based interference at some frequencies; the chosen tuning avoids the most persistent known RFI bands at the target redshift, and the 30% calibration/overhead allowance includes contingency for additional flagging [2].
- Scheduling: spectral-line HI observations are best scheduled at night to minimise solar interference; this is factored into the overhead estimate and does not otherwise constrain source visibility given the targets' declination range.

- Array readiness: this program is scoped for the AA* configuration (144 dishes), which is expected to be available for early science ahead of the completed AA4 array. The core sub-array adopted here (baselines ≤ 2 km) uses only already-commissioned SKA-Mid and MeerKAT dishes, so the quoted sensitivities do not depend on the array's outer spiral arms being complete. Should the array grow towards AA4 before scheduling, the same targets could be observed to correspondingly greater depth within an equivalent time allocation.

References

- [1] Wang et al. 2026, “HI Galaxy Science with the SKA”, in Advancing Astrophysics with the SKA – II (AASKAII).
- [2] de Blok et al. 2026, “Deep HI Observations of Cold Gas Inflow and Outflow”, AASKAII.
- [3] Ramatsoku et al. 2026, “Resolved HI and Environmental Dynamics”, AASKAII.
- [4] Ignesti et al. 2026, “Exploring the Physics of Ram Pressure Stripping with Radio Continuum Observations in the SKA Era”, AASKAII.
- [5] Serra et al. 2023, 2024, HI tidal tails in the Fornax Cluster, as summarised in Ramatsoku et al. 2026 (AASKAII).
- [6] Dewdney et al. 2022, “SKA1 Design Baseline Description”, SKA-TEL-SKO-0001075.
- [7] Seethapuram Sridhar et al. 2025, “SKAO Staged Delivery, Array Assemblies and Layouts”.
- [8] Nasirudin et al. 2026, “Cosmology from HI Galaxy Surveys with the SKA”, AASKAII.
- [9] Pan et al. 2026, “Studying HI and the Cosmic Web in the Era of SKA”, AASKAII.
- [10] Lelli et al. 2026, “Testing Dark Matter Models and Modified Gravity Theories with Spatially Resolved HI Observations”, AASKAII.
- [11] SKAO Sensitivity Calculator, sensitivity-calculator.skao.int (accessed August 2026); see also “Interacting with the SKAO Sensitivity Calculator” user guide.
- [12] Arumugam et al. 2025, “SKAO Science Data Products: a Summary”.